

Learning how to prove Workshop Week 10

Lagrange's Theorem



Cosets. Lagrange's Theorem.

1. Find the right cosets in G of the subgroup H of G .
 - (a) $G = C_{20} = \langle a \rangle$, $\text{o}(a) = 20$ and $H = \langle a^4 \rangle$.
 - (b) $G = \mathbb{Z}_{12}$, $H = \langle \bar{4} \rangle = \{\bar{0}, \bar{4}, \bar{8}\}$.
 - (c) Let G be a group and $a \in G$. Suppose that a has order 15. Find all the (distinct) right cosets of $\langle a^5 \rangle$ in $\langle a \rangle$.

Solution: (a) First of all notice that $H = \langle a^4 \rangle = \{1, a^4, a^8, a^{12}, a^{16}\}$.

$$\begin{aligned} H1 &= Ha^4 = Ha^8 = Ha^{12} = Ha^{16} = H \\ Ha &= \{a, a^5, a^9, a^{13}, a^{17}\} \\ Ha^2 &= \{a^2, a^6, a^{10}, a^{14}, a^{18}\} \\ Ha^3 &= \{a^3, a^7, a^{11}, a^{15}, a^{19}\} \end{aligned}$$

Taking into account that the right cosets form a partition of G , and that

$$G = H \cup Ha \cup Ha^2 \cup Ha^3,$$

we can conclude that the distinct right cosets of H are H, Ha, Ha^2, Ha^3 . If you keep computing cosets Ha^4, Ha^5 , etc. you will then get a repetition of the previous ones. For example:

$$Ha^5 = \{a^5, a^9, a^{13}, a^{17}, a^{21} = a\} = Ha$$

Another way to guarantee that those are all the different cosets of H in C_{20} is to find first the index of H in C_{20} . From $|C_{20} : H| = \frac{|C_{20}|}{|H|} = \frac{20}{5} = 4$, we derive that the number of different cosets of H in C_{20} is 4, and since H, Ha, Ha^2 and Ha^3 are all distinct, we can conclude that these are all.

(b) Recall that $\mathbb{Z}_{12}$ is an additive cyclic group, so $H = \langle \bar{4} \rangle = \{\bar{0}, \bar{4}, \bar{8}\} = 4\mathbb{Z}_{12}$.

$$\begin{aligned} H + \bar{0} &= H + \bar{4} = H + \bar{8} = H \\ H + \bar{1} &= \{\bar{1}, \bar{5}, \bar{9}\} \\ H + \bar{2} &= \{\bar{2}, \bar{6}, \bar{10}\} \\ H + \bar{3} &= \{\bar{3}, \bar{7}, \bar{11}\} \end{aligned}$$

Using again that the cosets $H + a$, $a \in \mathbb{Z}_{12}$ we can stop working because we have already listed all the elements of $\mathbb{Z}_{12}$. In other words, $H, H + \bar{1}, H + \bar{2}$ and $H + \bar{3}$ are the different cosets of $4\mathbb{Z}_{12}$ in $\mathbb{Z}_{12}$.

(c) Notice that $\langle a \rangle = \{1, a, a^2, \dots, a^{14}\}$ and $\langle a^5 \rangle = \{1, a^5, a^{10}\}$. The (distinct) right cosets of $\langle a^5 \rangle$ in $\langle a \rangle$ are:

$$\begin{aligned}\langle a^5 \rangle, \\ \langle a^5 \rangle a &= \{a, a^6, a^{11}\}, \\ \langle a^5 \rangle a^2 &= \{a^2, a^7, a^{12}\}, \\ \langle a^5 \rangle a^3 &= \{a^3, a^8, a^{13}\}, \\ \langle a^5 \rangle a^4 &= \{a^4, a^9, a^{14}\}.\end{aligned}$$

Taking into account that the right cosets form a partition of $\langle a \rangle$, and that

$$\langle a \rangle = \langle a^5 \rangle \cup \langle a^5 \rangle a \cup \langle a^5 \rangle a^2 \cup \langle a^5 \rangle a^3 \cup \langle a^5 \rangle a^4$$

we can conclude that these are all the distinct left cosets. If we keep computing cosets, we will then get a repetition of the previous ones. For example:

$$\langle a^5 \rangle a^5 = \{a^5, a^{10}, a^{15} = 1\} = H$$

Another way to guarantee that those are all the different cosets of $\langle a^5 \rangle$ in $\langle a \rangle$ is to find first the index of $\langle a^5 \rangle$ in $\langle a \rangle$. From $|\langle a \rangle : \langle a^5 \rangle| = \frac{|\langle a \rangle|}{|\langle a^5 \rangle|} = \frac{15}{3} = 5$, we derive that the number of different cosets is 5, so they must be the previous ones.

2. Let K and H subgroups of a group G and $g \in G$. Show that $g(H \cap K) = gH \cap gK$.

Solution: If $x \in g(H \cap K)$, then $x = ga$ for some $a \in H \cap K$. By definition, $ga \in gH$ and $ga \in gK$. This proves that $g(H \cap K) \subseteq gH \cap gK$.

Now, if $x \in gH \cap gK$, we have that $x = gh = gk$ for $h \in H$ and $k \in K$. By cancellation we get that $h = k$ and so $x \in g(H \cap K)$.